Homework 3

Expectation, variance, and joint discrete laws

60 points: 10 questions. Questions 4 and 7 are worth 10 points each; all other questions are worth 5 points each.

Due Tuesday, October 27, 2026, at 10 p.m. Pacific. The final no-penalty grace cutoff is Wednesday, October 28, at 10 p.m. Pacific. Approved accommodations apply.

Handwrite your work on paper or a tablet and submit one legible PDF in Gradescope, mapping pages to all ten questions, subject to approved accommodations. Show reasoning, support, and assumptions. Exact expressions are accepted unless stated otherwise. Write independently and disclose collaborators. Do not use AI tools to generate, solve, check, or rewrite assessed work.

Notation. PMF means probability mass function. $I_{\{C\}}$ is 1 when condition C holds and 0 otherwise. For a complete PMF, give every possible value and its probability, and state that the PMF is zero elsewhere. Tables, indicator formulas, and piecewise formulas are equally valid.

Work in stages. Questions 1–2 use the October 12 expectation lesson; 3–6 add October 19 variance; 7 adds October 21 joint laws; 8–10 add October 23 conditioning and independence. For course questions, use Ed Discussion or office hours.

1. Expectation of a discrete random variable

5 points

A random variable X has the following PMF, with zero mass elsewhere:

x	-3	1	4
$p_X(x)$	$1/4$	$1/2$	$1/4$

(a) [2] Compute $E[X]$ from the definition of expectation. Can X take the value $E[X]$? Explain how your answer relates to the meaning of expectation.

(b) [3] Let $Y = 2X + 1$. Compute $E[Y]$ using linearity of expectation. Check the result by calculating $E[Y]$ directly from the possible values of Y and their probabilities.

2. Drawing colored tokens

5 points

Draw four tokens in order, without replacement, from ten distinct tokens: four red and six blue. At each draw, every remaining token is equally likely to be selected. Let I_j equal 1 if draw j is red and 0 otherwise, for $j = 1, 2, 3, 4$. Let R be the number of red tokens among the four draws.

(a) [3] Express R in terms of I_1, I_2, I_3, I_4. Find $E[I_j]$ for each draw and use these expectations to find $E[R]$. State whether this calculation requires the indicators to be independent.

(b) [2] Find $P(I_2 = 1 \mid I_1 = 1)$ and $P(I_2 = 1)$. Use the comparison to decide whether I_1 and I_2 are independent. Does your conclusion affect the validity of part (a)? Explain.

3. The square of a discrete random variable

5 points

Each value in $\{-2, -1, 0, 1, 2\}$ has probability $1/5$ for X. Define $Z = X^2$.

(a) [3] Find the complete PMF of Z and then compute $E[Z]$ from that PMF.

(b) [2] Compute $E[X]$ and $E[X^2]$ directly from the PMF of X. Is $E[X^2] = (E[X])^2$ true for this distribution? Justify your answer.

4. Mean and variability of a delay

10 points

A task has additional delay D, measured in milliseconds, with

d	0	2	6
$p_D(d)$	1/2	1/3	1/6

The PMF is zero elsewhere.

(a) [5] Compute $E[D]$, $E[D^2]$, $\mathrm{Var}(D)$, and $\mathrm{SD}(D)$, using the second-moment formula for variance. State the units of all four quantities.

(b) [5] Check your variance by evaluating $E[(D - E[D])^2]$ directly from the PMF. Explain whether the variance is positive. Also evaluate $E[D - E[D]]$ and explain whether it can measure variability.

5. Recalibrating a measurement

5 points

A measurement X has mean 12 and variance 4. Define the recalibrated reading $Y = 50 - 3X$.

(a) [3] Find $E[Y]$, $\mathrm{Var}(Y)$, and $\mathrm{SD}(Y)$. Explain how the coefficient -3 affects the variance and standard deviation.

(b) [2] A student claims $\mathrm{Var}(50 - 3X) = 50 - 3\,\mathrm{Var}(X)$. State the correct general rule for $\mathrm{Var}(aX + b)$, where a, b are constants. If Y is replaced by $Y^* = 55 - 3X$, which quantities in part (a) change? Justify your answer by comparing $Y - E[Y]$ with $Y^* - E[Y^*]$.

6. Mean and variance of an indicator

5 points

Let I indicate an event with probability p, where $0 \le p \le 1$. Thus $P(I = 1) = p$ and $P(I = 0) = 1 - p$.

(a) [3] Compute $E[I]$ and $E[I^2]$ from the two possible values of I and their probabilities. Use these results to derive $\mathrm{Var}(I)$.

(b) [2] Calculate the variance when $p = 1/5$ and when $p = 1/2$. Which has greater variability, as measured by variance? Find the variance for $p = 0$ and for $p = 1$, and explain these two cases.

7. A joint PMF and the sum of two variables

10 points

The joint PMF of X, Y is given by the table below and is zero for all other pairs. The constant c makes it a valid PMF.

$X \backslash Y$	0	1
0	c	$2c$
1	$2c$	c
2	0	$4c$

(a) [5] Find c and the complete marginal PMFs $p_X(x)$ and $p_Y(y)$. Verify that the joint probabilities sum to 1 and that each marginal PMF also sums to 1.

(b) [5] Define $S = X + Y$. Find the complete PMF of S and compute $E[S]$ from it. Check the result by calculating $E[X] + E[Y]$ from the marginal PMFs.

8. Conditional probabilities from a joint PMF

5 points

Use the joint PMF in Question 7.

(a) [3] Find the complete conditional PMF $p_{Y|X}(y \mid 1)$ and the probability $P(X = 1 \mid Y = 1)$. Show the conditional-probability calculation for each.

(b) [2] Find the complete conditional PMF $p_{X|Y}(x \mid 0)$ and verify that it sums to 1. Is $P(Y = 1 \mid X = 3)$ defined by the elementary definition of conditional probability? Explain.

9. Comparing two joint distributions

5 points

Use the marginal PMFs from Question 7. Let U and V be independent, with $p_U(u) = p_X(u)$ for every u and $p_V(v) = p_Y(v)$ for every v. Thus U has the same marginal distribution as X, and V has the same marginal distribution as Y.

(a) [3] Are the original variables X and Y independent? Justify your answer with a probability comparison. Construct the complete joint PMF table for U and V.

(b) [2] Calculate and compare $P(X + Y = 3)$ and $P(U + V = 3)$. Then calculate and compare $E[X + Y]$ and $E[U + V]$. Explain what the two comparisons show about the role of the joint distribution.

10. A joint PMF from conditional probabilities

5 points

The random variable A takes values 0 and 1, with $P(A = 0) = 2/5$ and $P(A = 1) = 3/5$. For each possible value of A, the conditional PMF of B is given below; it is zero for all other b:

b	0	1	2
$p_{B\|A}(b \mid 0)$	1/2	1/4	1/4
$p_{B\|A}(b \mid 1)$	1/6	1/2	1/3

(a) [3] Find the complete joint PMF $p_{A,B}(a, b)$ and the complete marginal PMF $p_B(b)$. Verify that each PMF sums to 1.

(b) [2] Find $P(A = 0 \mid B = 2)$. Are A and B independent? Justify your conclusion with a probability comparison.

Reading references. Harchol-Balter: pp. 58–64 (expectation), 83–88 (moments and variance), 50–51 and 67–68 (joint and conditional PMFs). DeGroot–Schervish: Sections 4.1–4.3 and 3.4–3.6. Printed book pages; consult selected passages as needed. Neither whole chapters nor both books are required.